

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

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11 Watchdog PCB external connectors

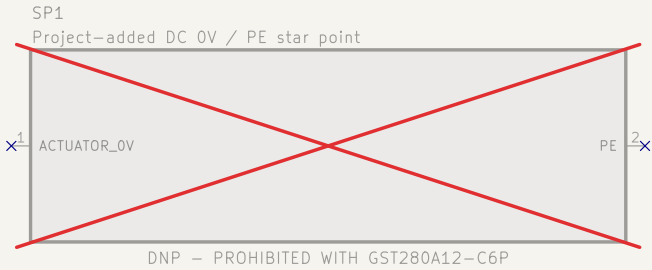
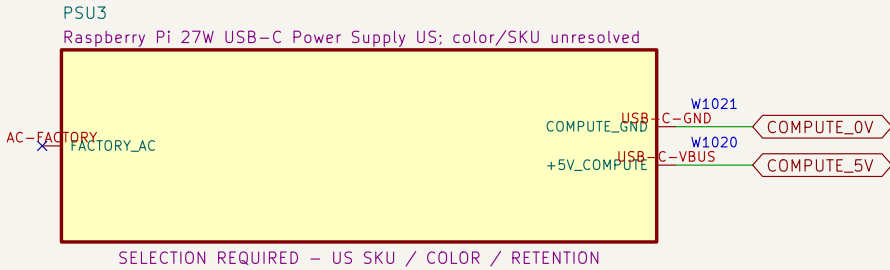
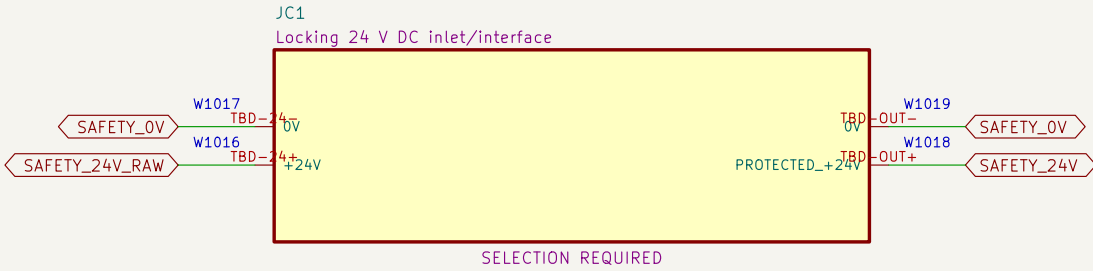
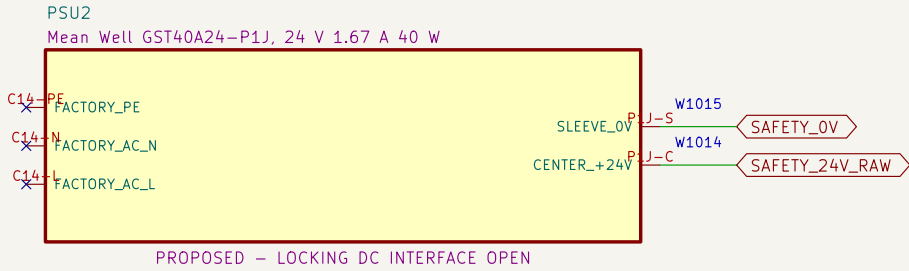
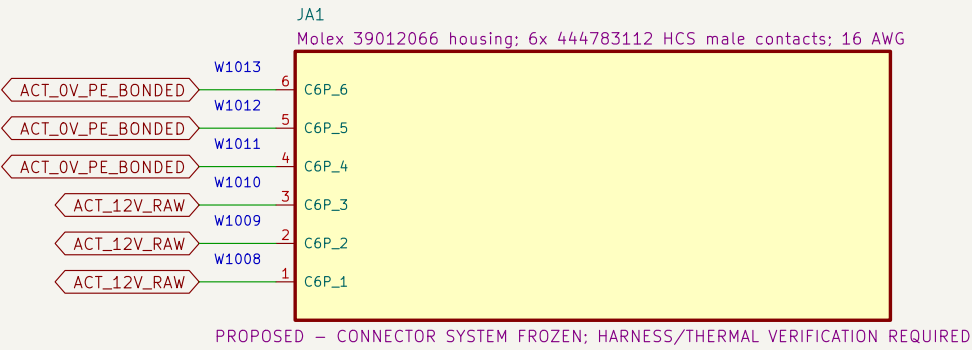
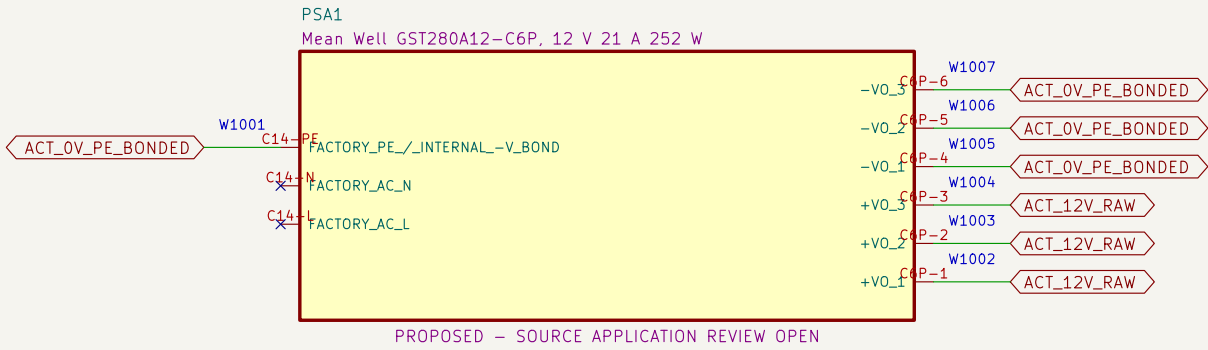
File: 11_watchdog_pcb_connectors.kicad_sch

12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch

01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.

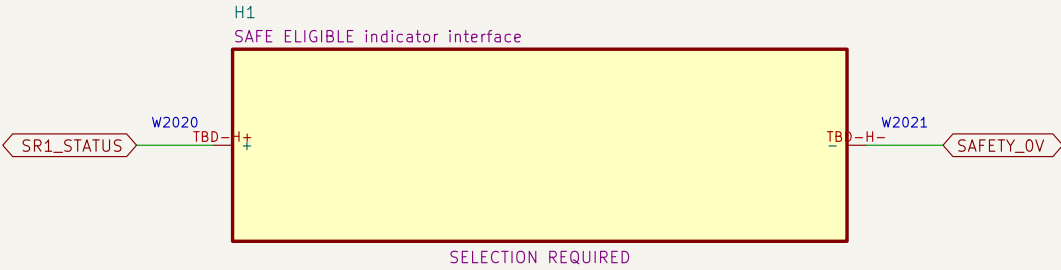
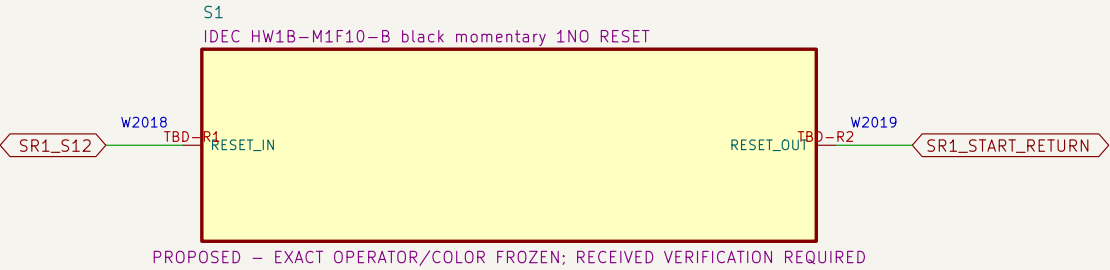
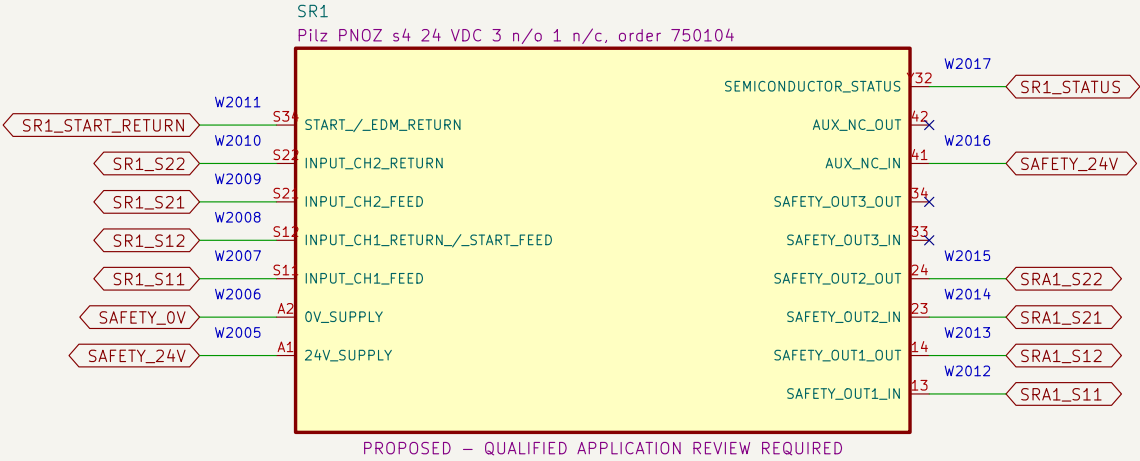
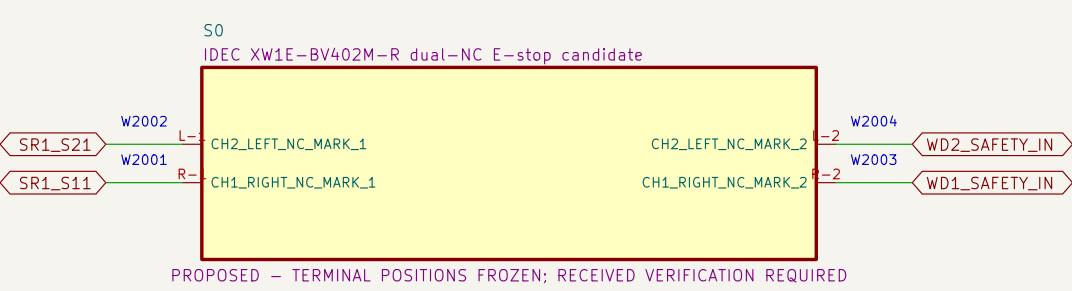


NOTE 1: All AC conductors shown are factory boundaries, not project-built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains one E-stop NC and one watchdog NO contact; RESET cannot energize K1/K2.

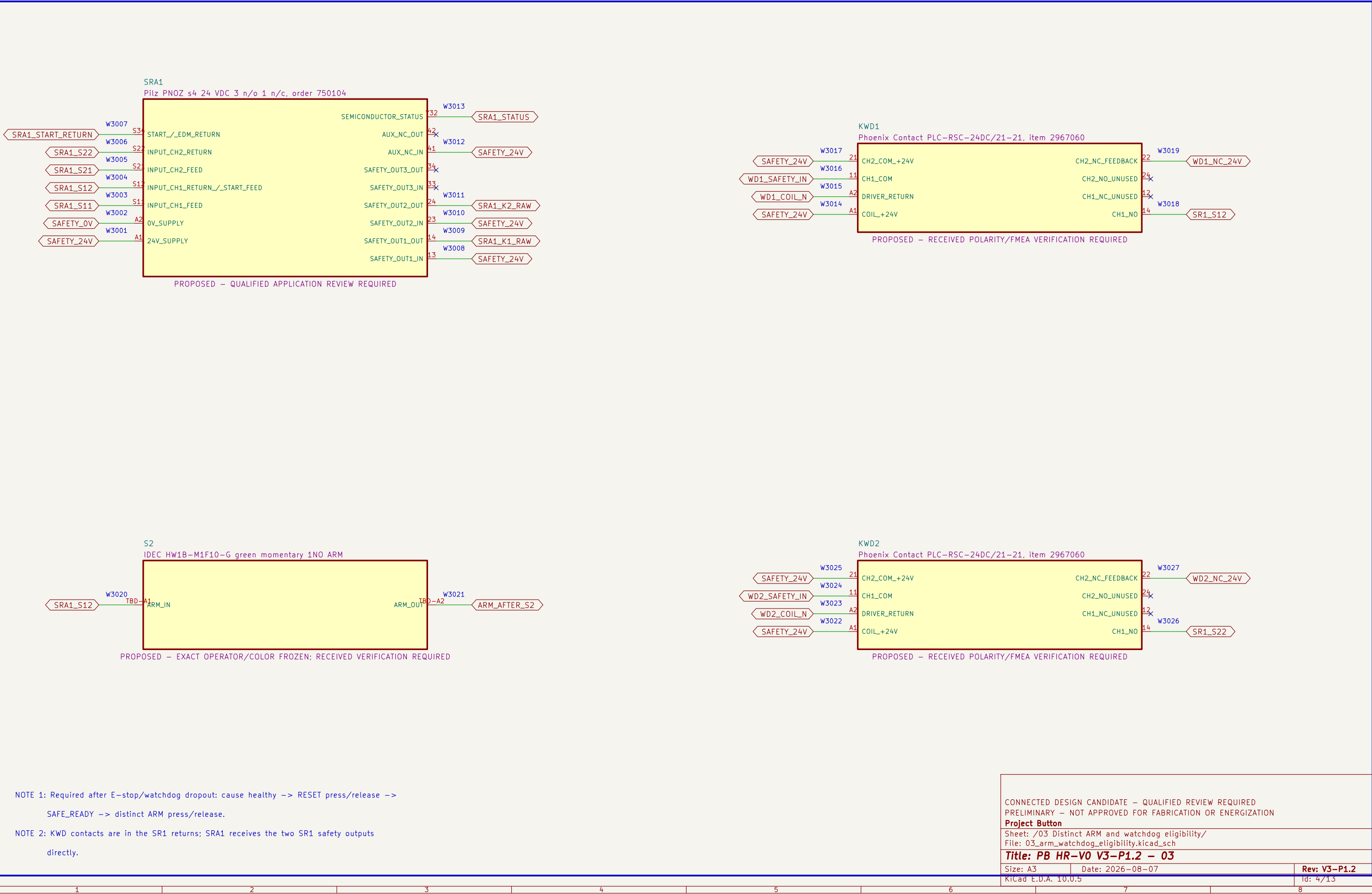


NOTE 1: Heartbeat loss opens both SR1 input returns; recovery alone cannot restore the monitored RESET stage.

NOTE 2: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized until a later ARM.

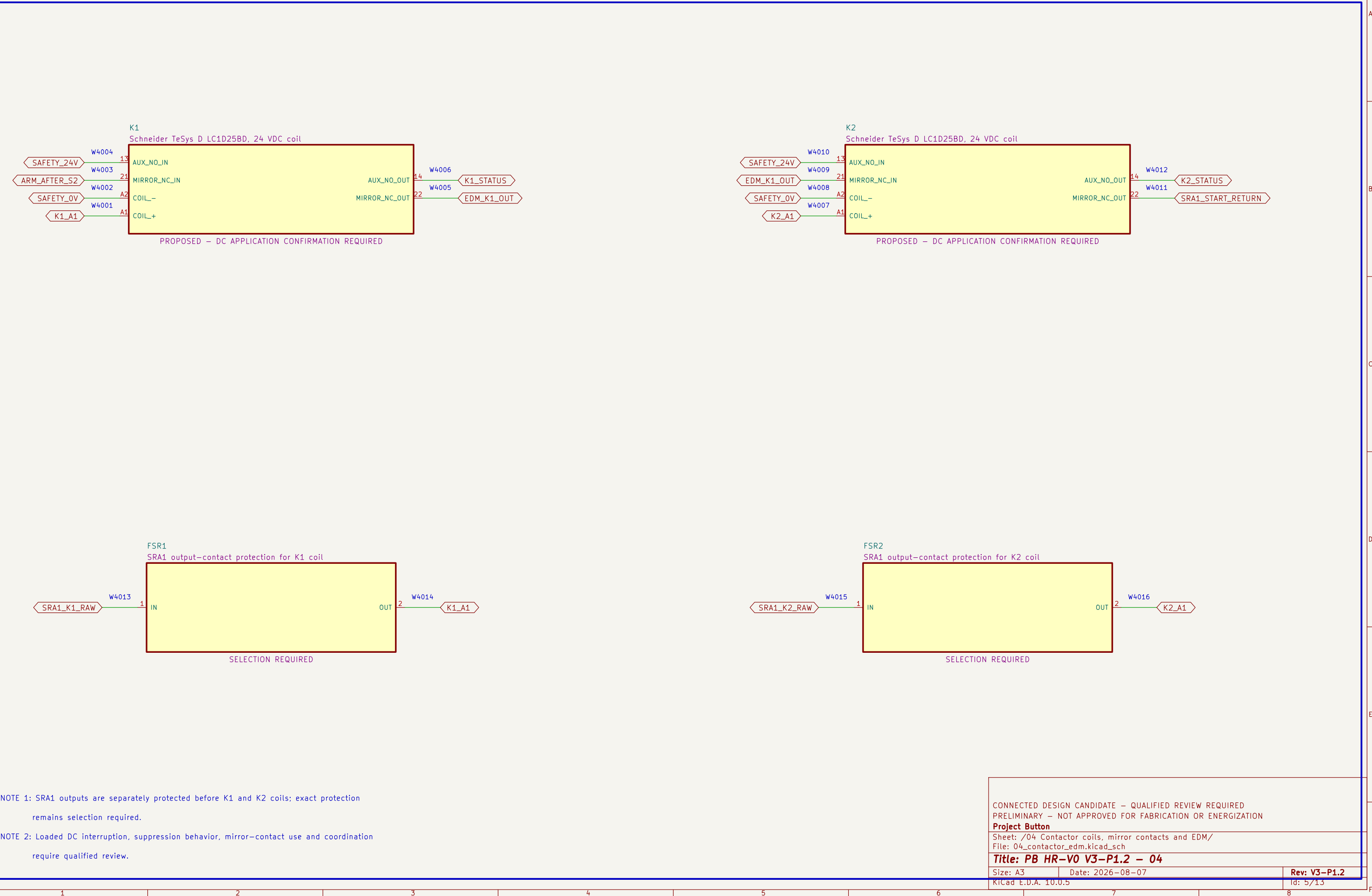
03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.



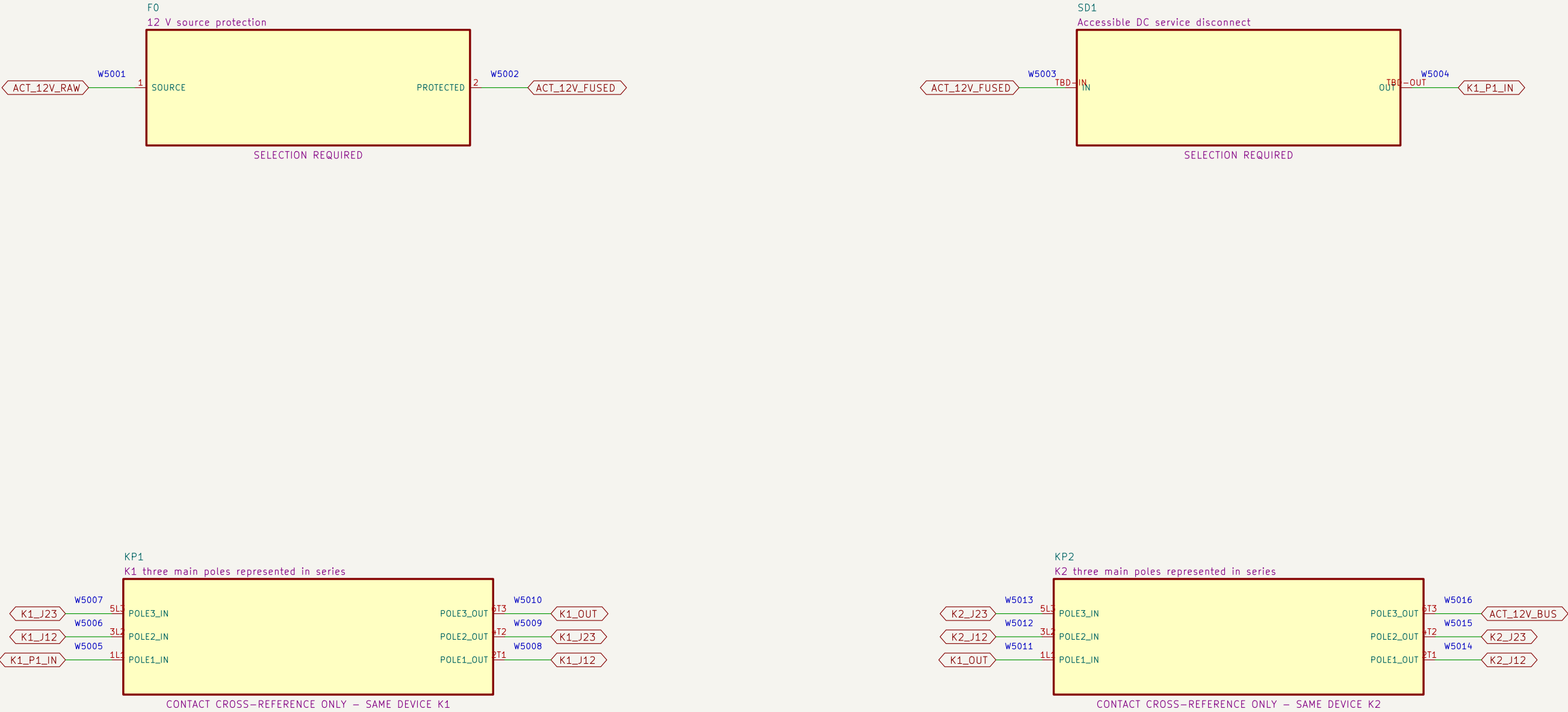
04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



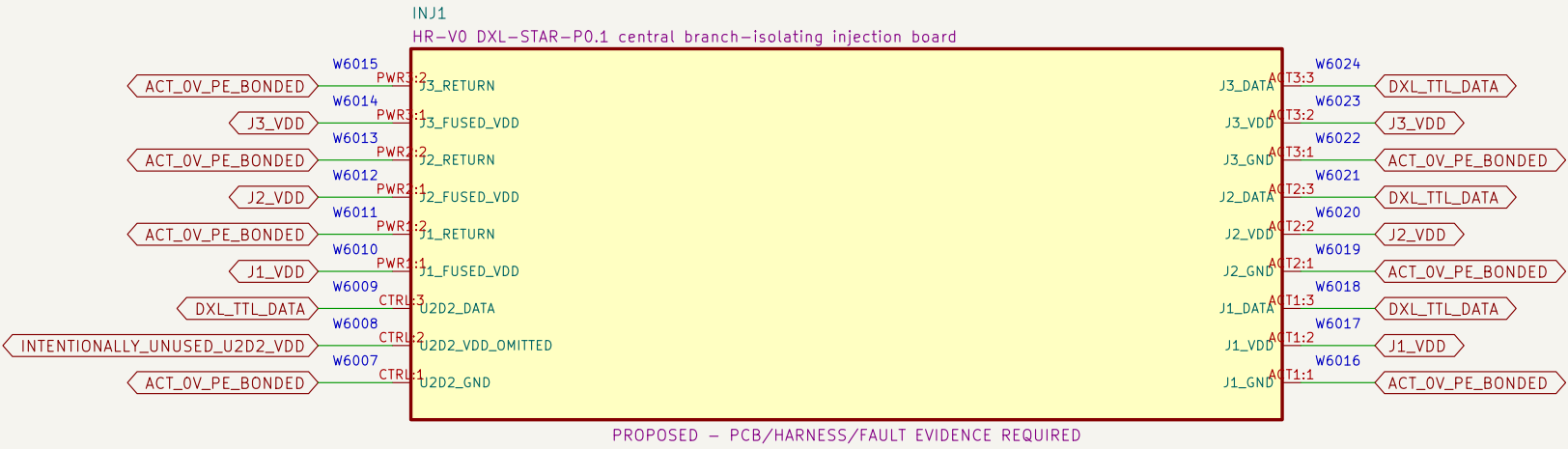
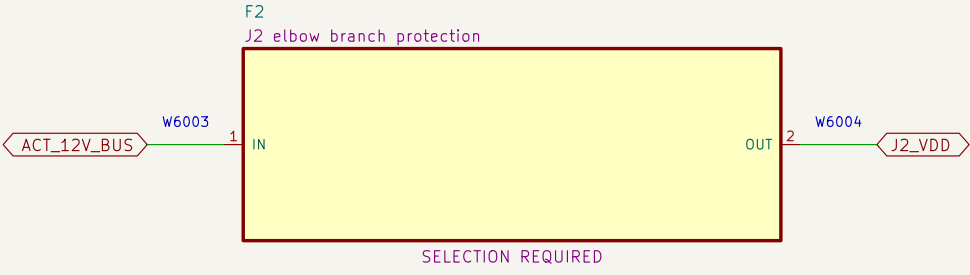
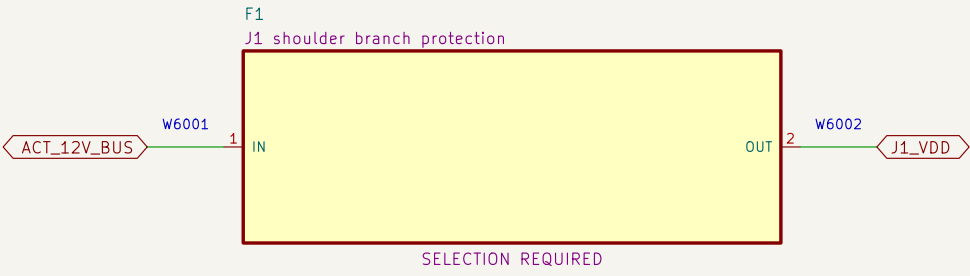
NOTE 1: Series jumpers: K1 2T1→3L2, 4T2→5L3, 6T3→K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /05 Redundant actuator–power interruption/
File: 05_actuator_interruption.kicad_sch
Title: PB HR–V0 V3–P1.2 – 05
Size: A3 Date: 2026–08–07 **Rev: V3–P1.2**
KiCad E.D.A. 10.0.5 Id: 6/13

06 Protected actuator branches and central DYNAMIXEL star injection

Each actuator has a separate protected VDD branch; U2D2 pin 2 and inter-actuator VDD paths are omitted.



NOTE 1: INJ1 is a fixed central star; U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

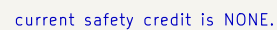
NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal-integrity, pull and no-backfeed tests remain release gates.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button
Sheet: /06 Protected actuator branches and central DYNAMIXEL star injection/
File: 06_branches_and_injection.kicad_sch

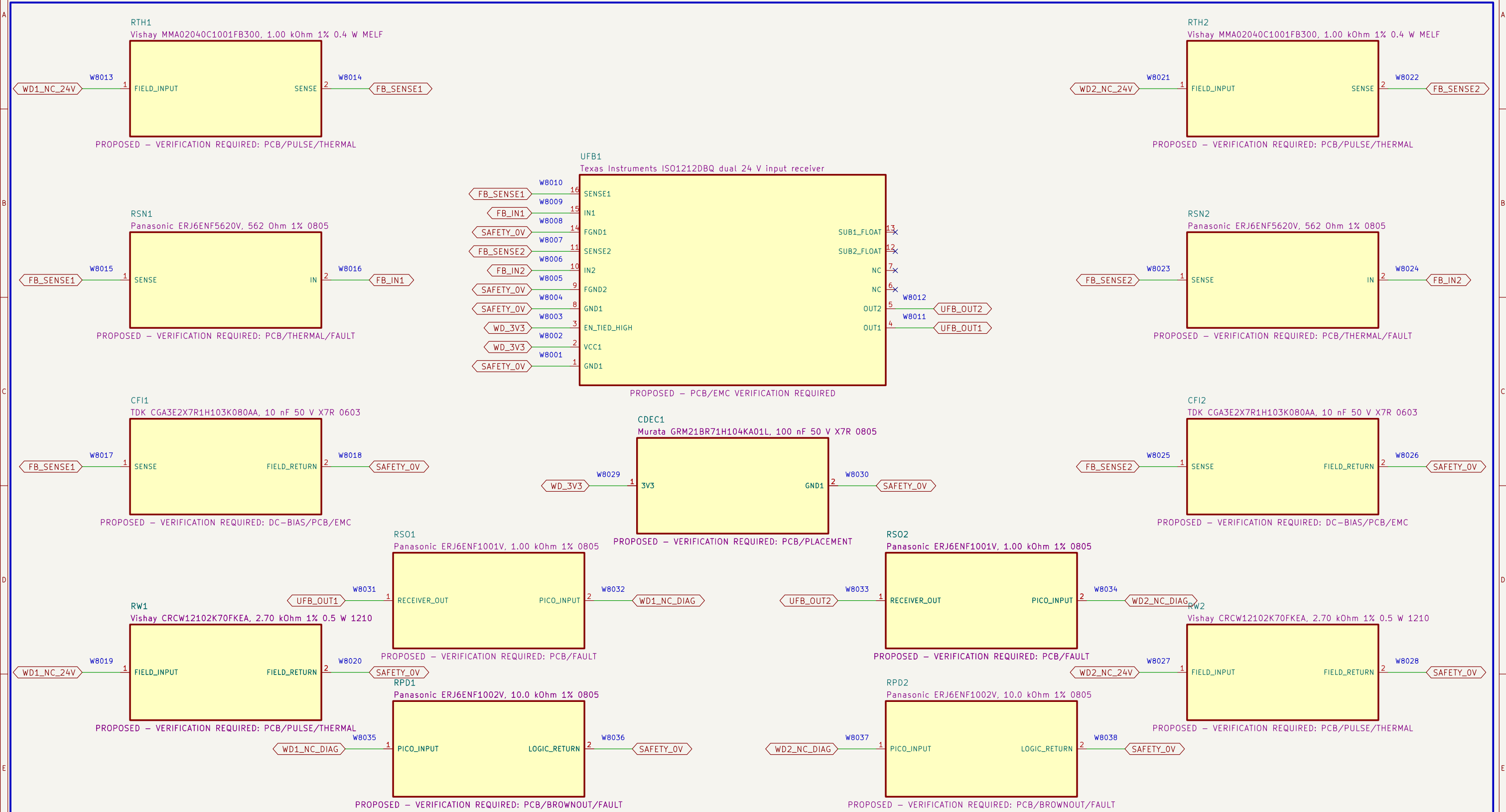
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Size: A3	Date: 2026-08-07	Rev: V3-P1.2
KiCad E.D.A. 10.0.5		Id: 7/13



08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



NOTE 1: Type-3 values: RTHR=1.00 kOhm, RSENSE=562 Ohm, CIN=10 nF; calculated wetting load is 2.70 kOhm 1% 0.5 W per channel.

NOTE 2: Both grounds are SAFETY_OV; the ISO1212 barrier is not credited. Passive identities are frozen; PCB, derating, EMC and fault tests remain open.

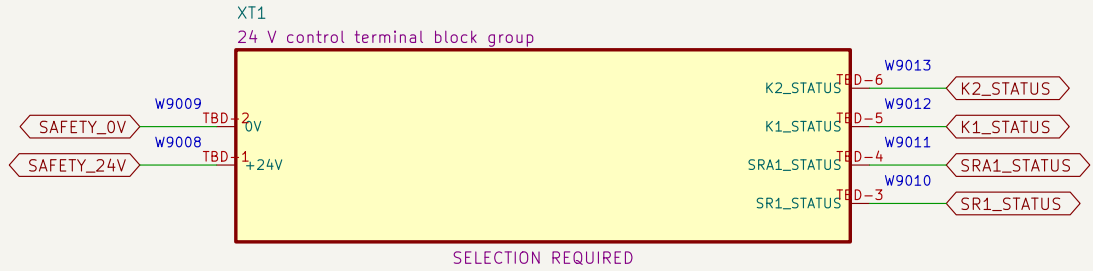
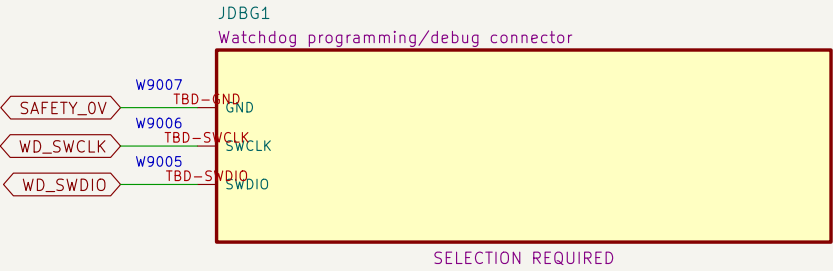
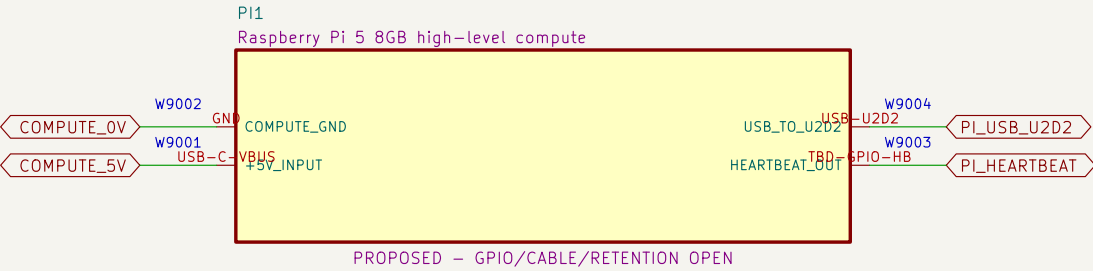
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /08 Calculated dual-channel 24 V watchdog feedback/
File: 08_watchdog_feedback_interface.kicad_sch

Title: PB HR-V0 V3-P1.2 – 08

Size: A3	Date: 2026-08-07	Rev: V3-P1.2
KiCad E.D.A. 10.0.5		Id: 9/13

09 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.

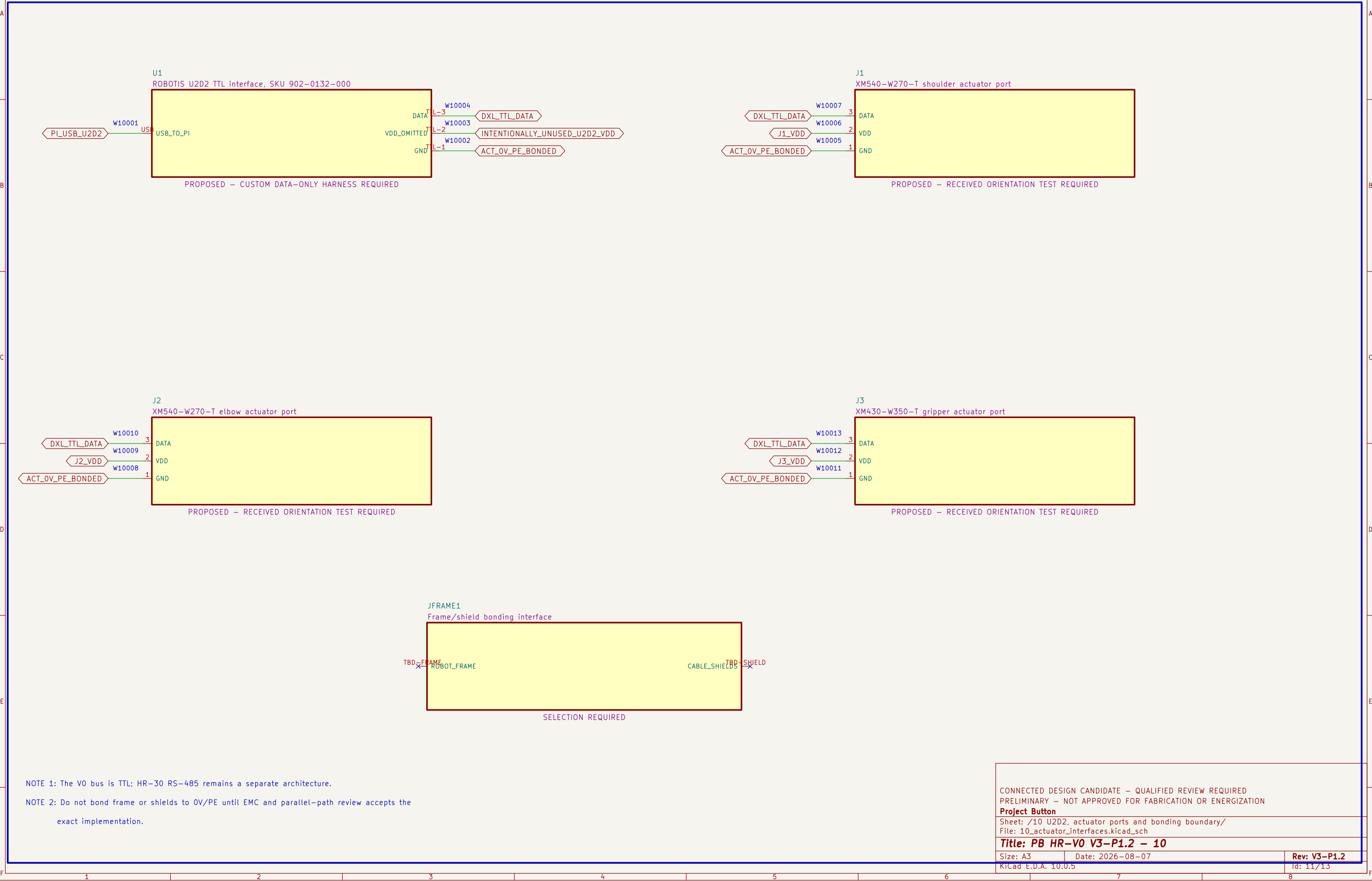


NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

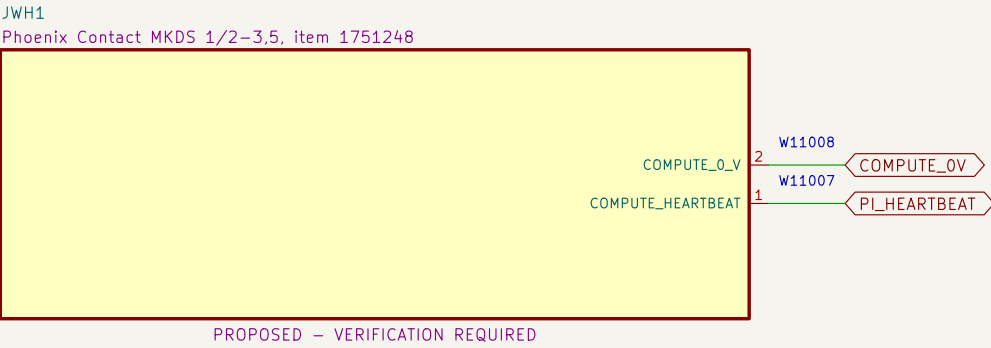
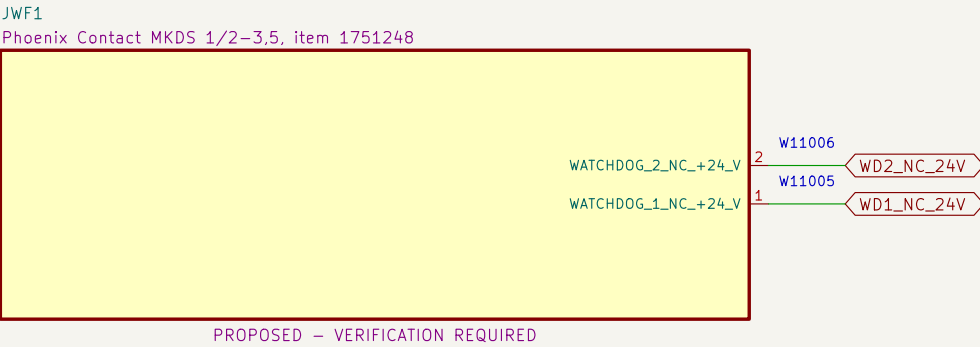
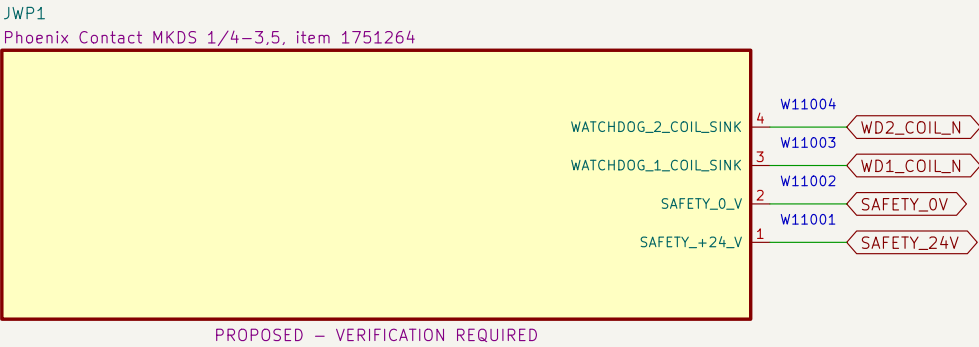
10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



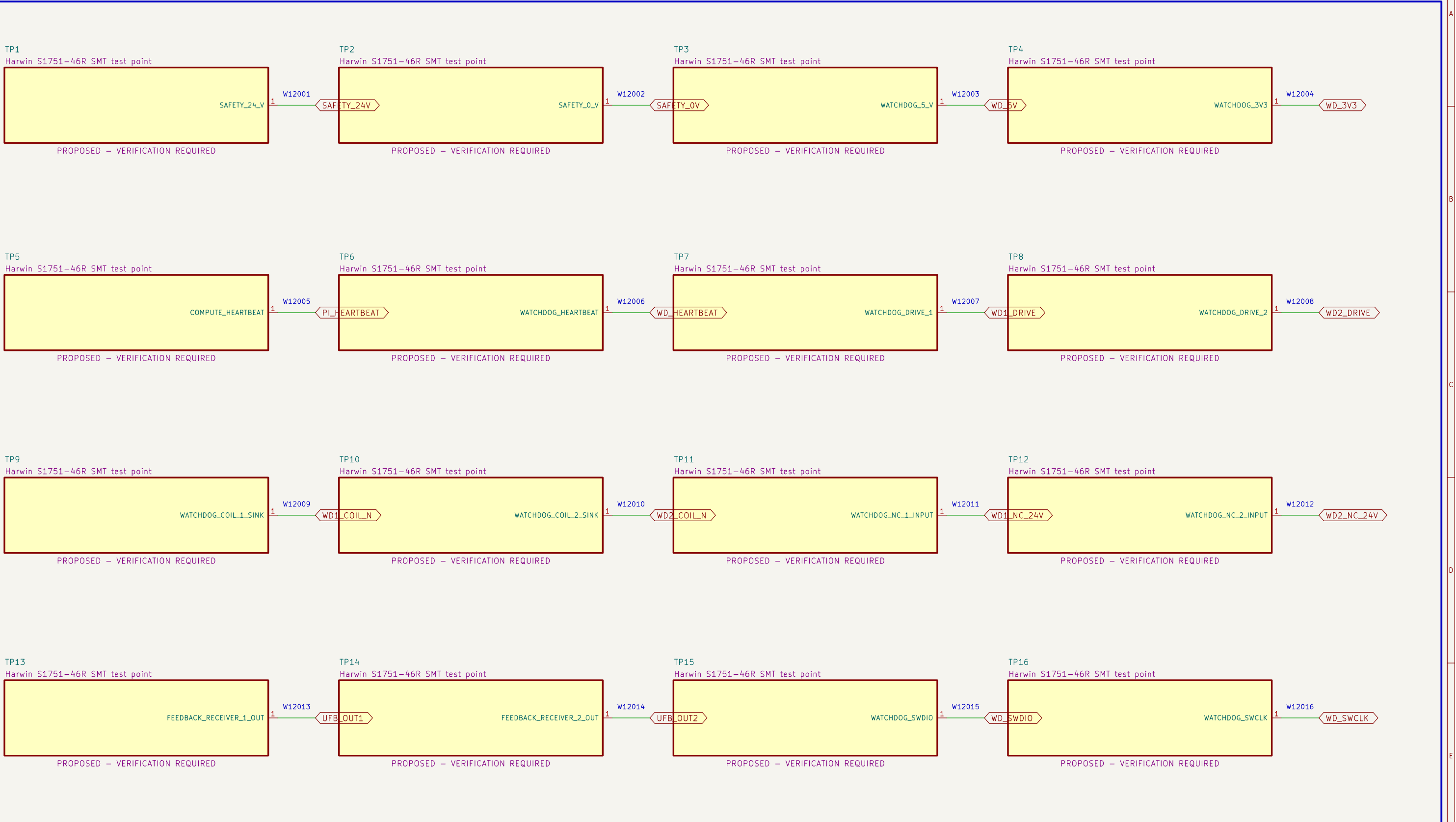
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION Project Button		
Sheet: /11 Watchdog PCB external connectors/ File: 11_watchdog_pcb_connectors.kicad_sch		
Title: PB HR–V0 V3–P1.2 – 11		
Size: A3	Date: 2026–08–07	Rev: V3–P1.2
KiCad E.D.A. 10.0.5		Id: 12/13

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /12 Watchdog PCB test access/		
File: 12_watchdog_pcb_test_access.kicad_sch		
Title: PB HR-V0 V3-P1.2 – 12		
Size: A3	Date: 2026-08-07	Rev: V3-P1.2
KiCad E.D.A. 10.0.5		Id: 13/13